RTCA, Inc.

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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 19

Induced Signal Susceptibility

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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19.0 Induced Signal Susceptibility

19.1 Purpose of the Test

This test determines whether the equipment interconnect circuit configuration will accept a level of induced voltages caused by the installation environment. This section relates specifically to interfering signals related to the power frequency and its harmonics, audio frequency signals, and electrical transients that are generated by other on-board equipment or systems and coupled to sensitive circuits within the EUT through its interconnecting wiring. Power leads tested under Section 18 are exempt from this test.

19.2 Equipment Categories

The induced signal susceptibility tests category is defined with three letters. The first letter (C, Z, A or B) refers to the test level; the second letter (C, N or W) refers to the type of primary power supply installation; and the third letter (E or X) indicates whether equipment is qualified to "E-field into equipment" (paragraph 19.3.2) tests (E) or not (X).

Category C_
Equipment intended primarily for operation in systems where interference-free operation is required and where severe coupling occurs due to long wire runs or minimum wire separation is identified as Category C_.

Category CC: Equipment is installed on aircraft whose primary power is constant frequency (e.g. 400 Hz) or DC

Category CN: Equipment is installed on aircraft whose primary power is variable over a narrow frequency range (e.g. 350 Hz – 650 Hz)

Category CW: Equipment is installed on aircraft whose primary power is variable over a wide frequency range (e.g. 350 Hz – 800 Hz)

Category Z_
Equipment intended primarily for operation in systems where interference-free operation is required is identified as Category Z_.

Category ZC: Equipment is installed on aircraft whose primary power is constant frequency (e.g. 400 Hz) or DC

Category ZN: Equipment is installed on aircraft whose primary power is variable over a narrow frequency range (e.g. 350 Hz – 650 Hz)

Category ZW: Equipment is installed on aircraft whose primary power is variable over a wide frequency range (e.g. 350 Hz – 800 Hz)

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Category A_
Equipment intended primarily for operation where interference-free operation is desirable is identified as Category A_.

Category AC: Equipment is installed on aircraft whose primary power is constant frequency (e.g. 400 Hz) or DC

Category AN: Equipment is installed on aircraft whose primary power is variable over a narrow frequency range (e.g. 350 Hz – 650 Hz)

Category AW: Equipment is installed on aircraft whose primary power is variable over a wide frequency range (e.g. 350 Hz – 800 Hz)

Category B_
Equipment intended primarily for operation in systems where interference would be controlled to a tolerable level is identified as Category B_.

Category BC: Equipment is installed on aircraft whose primary power is constant frequency (e.g. 400 Hz) or DC

Category BN: Equipment is installed on aircraft whose primary power is variable over a narrow frequency range (e.g. 350 Hz – 650 Hz)

Category BW: Equipment is installed on aircraft whose primary power is variable over a wide frequency range (e.g. 350 Hz – 800 Hz)

19.3 Test Procedures

19.3.1 Magnetic Fields Induced Into the Equipment

Subject the equipment under test to an audio frequency magnetic field, generated by the current specified in Table 19-1, in a straight wire radiator located within 0.15 m of the periphery of the unit of equipment under test. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

During this test, the radiating wire shall be oriented with respect to each external surface of each unit to cause maximum interference. The length of the radiator shall extend a distance of at least 0.6 m (laterally) beyond each extremity of the unit under test. The leads applying current to the radiator shall be routed at least 0.6 m away from any part of the unit under test and from the radiator itself. All units of the equipment under test shall be individually tested. The magnetic field power source shall not be synchronized with the power source of the equipment supply. During this test, place the wire horizontally and sweep up the front face of the EUT, across the top and down the back face of the EUT. Turn the radiating wire or EUT 90 degrees and sweep up the left (or right) face of the EUT, across the top, and down the right (or left) face of the EUT. Additional

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radiating wire/EUT orientations are permissible. The goal is to test all four sides and the top of the box. Sweep at a rate slower than the system response time.

19.3.2 Electric Fields Induced Into the Equipment

Subject the equipment under test to an audio frequency electric field generated by 170V rms in a straight wire radiator of 0.20 m length located within 0.01 m of the periphery of the unit of equipment under test. See Table 19-1 for applicable test frequencies and levels. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

During this test, the radiator shall be oriented with respect to each external surface of each unit to cause maximum interference. For setup see Fig. 19-2. The leads applying voltage to the radiator shall be routed at least 0.6 m away from any part of the unit under test and from the radiator itself. All units of the equipment under test shall be individually tested. The electric field power source shall not be synchronized with the power source of the equipment supply.

During this test, place the wire horizontally and sweep up the front face of the EUT, across the top and down the back face of the EUT. Turn the radiating wire or EUT 90 degrees and sweep up the left (or right) face of the EUT, across the top, and down the right (or left) face of the EUT. The goal is to test all four sides and the top of the box. Sweep at a rate slower than the system response time.

A resistor sized for personnel protection shall be installed in the radiator wire.

Note: The Electric Field test should be applied only to equipment which is intended for installation without external metallic containment, or which is covered by an enclosure made from non-conducting material.

19.3.3 Magnetic Fields Induced Into Interconnecting Cables

Subject the interconnecting wire bundle of the equipment under test to an audio frequency magnetic field as illustrated in Figure 19-3. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS when the field is of the value specified in Table 19-1.

During this test, all equipment interconnecting cables shall be installed in accordance with the applicable installation and interface control diagrams. Any inputs or outputs from or to other equipment(s) normally associated with the equipment under test shall be adequately simulated. The magnetic field power source is not to be synchronized with the power source of the equipment power supply. Frequency scan rates and dwell times shall be in accordance with section 19.3.6

19.3.4 Electric Fields Induced Into Interconnecting Cables

Subject the interconnecting wire bundle of the equipment under test to an audio frequency electric field as illustrated by Figure 19-4. DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS when the field is of the value specified in Table 19-1.

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During this test, all equipment interconnecting cables shall be in accordance with the applicable installation and interface control diagrams. Shielded or twisted wires shall be used only where specified by the equipment manufacturer. Any inputs or outputs from or to other equipment(s) normally associated with the equipment under test shall be adequately simulated. The electric field power source shall not be synchronized with the power source of the equipment power supply. Frequency scan rates and dwell times shall be in accordance with section 19.3.6.

Note: Spiral wrapping of the radiator wire around the wire bundle under test is required for this test. The radiator wire shall be spiral wrapped around the bundle under test using 3 spiral wraps per meter length of the bundle, along the entire length L, as defined in Figure 19-4

19.3.5 Spikes Induced Into Interconnecting Cables

During this test, all equipment interconnecting cables shall be in accordance with the applicable installation and interface control diagrams. Shielded or twisted wires shall be used only where specified by the equipment manufacturer. Any inputs or outputs from or to other equipment normally associated with the equipment under test shall be adequately simulated.

Subject the interconnecting wire bundle of the equipment under test to both positive and negative transient fields using the test setup shown in Figure 19-5. Table 19-1 defines the desired cable lengths for categories of section 19.2. The timer shown in Figure 19-5 shall be adjusted to yield a pulse repetition rate of eight to ten pulses each second. The waveform present at point A, Figure 19-5, should be similar to that described in Figure 19-6. For both positive and negative polarities of the transient, the pulsing for each polarity shall be maintained for a period of not less than two minutes or for a longer period of time if specified in the relevant equipment specification.

Note: Spiral wrapping of the radiator wire around the wire bundle under test is required for this test. The radiator wire shall be spiral wrapped around the bundle under test using 3 spiral wraps per meter length of the bundle, along the entire length L, as defined in Figure 19-5

After exposure, DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS. Any requirement for performance of the equipment during application of the tests will be specified in the equipment performance standard.

The inductive switching transient generated when the contact opens should be very similar to the illustration in Figure 19-6, when monitored at point A on Figure 19-5. When the contact opens, the voltage at Point A drives from +28 V dc to large negative voltages in about two microseconds. (The capacitance, 250 to 3,000 picofarad typically, between windings of the coil is charged negatively during this time.) When the voltage reaches the ionizing potential, arc-over occurs at the contact and the voltage drives rapidly toward 28 V dc through the ionized path at the contact. The voltage at Point A

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usually overshoots +28 V dc because of the wire inductance between Point A and the coil. At this time, the arc extinguishes and the cycle is repeated. In a typical case, the repetition period is 0.2 to 10 microseconds and the number of repetitions is often 5 to 1,000 before the energy of the inductive load ($E = 1/2\ LI^2$) is dissipated.

19.3.6 Frequency Scan Rates

For test equipment that generate discrete frequencies, the minimum number of test frequencies shall be 30 frequencies per decade. The test frequencies shall be logarithmically spaced. As an example a formula which can be used to calculate these frequencies for 30 steps per decade in ascending order is:

$$f_{n+1} = f_n * 10^{(1/30)}$$

where f_n is a test frequency and $n = 1$ to m,
f_1 is the start frequency
f_m is the end frequency
$m = 1+30*\log(f_m/f_1)$

The dwell time at each test frequency shall be at least 10 seconds, exclusive of test equipment settling time. If the last test step for f_{n+1} yields a frequency greater than f_m, round down to f_m.

For test equipment that generate a continuous frequency sweep, the minimum (i.e., fastest) sweep rate shall be equal to the number of discrete frequencies per decade (m) multiplied by the dwell time, i.e., 30 discrete frequencies per decade times 10 second dwell time equals 5 minutes.

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Table 19-1 Applicability of Categories to Induced Signal Susceptibility

Paragraph	Test	Category ZC	Category AC	Category BC	Category CC
19.3.1	Magnetic Fields induced into the equipment	20 A rms at 400 Hz	20 A rms at 400 Hz	20 A rms at 400 Hz	20 A rms at 400 Hz
19.3.2	Electric Fields induced into the equipment	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz
19.3.3	Magnetic fields induced into interconnecting cables	IxL=30 A-m at 400 Hz reducing to 0.8 A-m at 15 kHz	IxL=18 A-m from 380 to 420 Hz	Not Applicable	IxL=120 A-m from 380 to 420 Hz and 60 A-m at 400 Hz reducing to 1.6 A-m at 15 kHz
19.3.4	Electric Fields induced into interconnecting cables	VxL=1800 V-m from 380 to 420 Hz	VxL=360 V-m from 380 to 420 Hz	Not Applicable	VxL=5400 V-m from 380 to 420 Hz and 5400 V-m at 400 Hz reducing to 135 V-m at 15 kHz
19.3.5	Spikes induced into interconnecting cables	Figure 19-4 L=3.0 m	Figure 19-4 L=3.0 m	Figure 19-4 L=1.2 m	Figure 19-4 L=3.0 m

Paragraph	Test	Category ZN	Category AN	Category BN	Category CN
19.3.1	Magnetic Fields induced into the equipment	20 A rms at 350 Hz and 650 Hz	20 A rms at 350 Hz and 650 Hz	20 A rms at 350 Hz and 650 Hz	20 A rms at 350 Hz and 650 Hz
19.3.2	Electric Fields induced into the equipment	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz
19.3.3	Magnetic fields induced into interconnecting cables	IxL=30 A-m from 350 to 650 Hz and reducing to 0.8 A-m at 26 kHz	IxL=18 A-m from 350 to 650 Hz	Not Applicable	IxL=120 A-m from 350 to 650 Hz reducing to 1.6 A-m at 26 kHz
19.3.4	Electric Fields induced into interconnecting cables	VxL=1800 V-m from 350 to 650 Hz	VxL=360 V-m from 350 to 650 Hz	Not Applicable	VxL=5400 V-m from 350 to 650 Hz reducing to 135 V-m at 26 kHz
19.3.5	Spikes induced into interconnecting cables	Figure 19-4 L=3.0 m	Figure 19-4 L=3.0 m	Figure 19-4 L=1.2 m	Figure 19-4 L=3.0 m

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Table 19-1 Applicability of Categories to Induced Signal Susceptibility (continued)

Paragraph	Test	Category ZW	Category AW	Category BW	Category CW
19.3.1	Magnetic Fields induced into the equipment	20 A rms at 350 Hz and 800 Hz	20 A rms at 350 Hz and 800 Hz	20 A rms at 350 Hz and 800 Hz	20 A rms at 350 Hz and 800 Hz
19.3.2	Electric Fields induced into the equipment	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz	170 V rms at 400 Hz
19.3.3	Magnetic fields induced into interconnecting cables	IxL=30 A-m from 350 to 800 Hz and reducing to 0.8 A-m at 32 kHz	IxL=18 A-m from 350 to 800 Hz	Not Applicable	IxL=120 A-m from 350 to 800 Hz reducing to 1.6 A-m at 32 kHz
19.3.4	Electric Fields induced into interconnecting cables	VxL=1800 V-m from 350 to 800 Hz	VxL=360 V-m from 350 to 800 Hz	Not Applicable	VxL=5400 V-m from 350 to 800 Hz reducing to 135 V-m at 32 kHz
19.3.5	Spikes induced into interconnecting cables	Figure 19-4 L=3.0 m	Figure 19-4 L=3.0 m	Figure 19-4 L=1.2 m	Figure 19-4 L=3.0 m

Note: When the manufacturer's installation and interface control drawings or diagrams specify a fixed length cable less than 3.3 m, the coupled length (L) of the field source wire shall be reduced to maintain the 0.15 m minimum separation distance at each end. The test level may be adjusted downward in proportion to the ratio of the reduced coupling length to the specified coupling length (L = 3.3 - (2 x 0.15) = 3 m). For example, if the manufacturer specifies a maximum cable length of 1.8 m, then L = 1.5 m. The adjusted level is then (1.5/3.0) = 0.5 times the voltage (V-m) or current (A-m) test limts shown above. This test is not required when the manufacturer's installation and interface control drawings or diagrams specify a fixed length cable less than 1.5 m (L less than 1.2 m).

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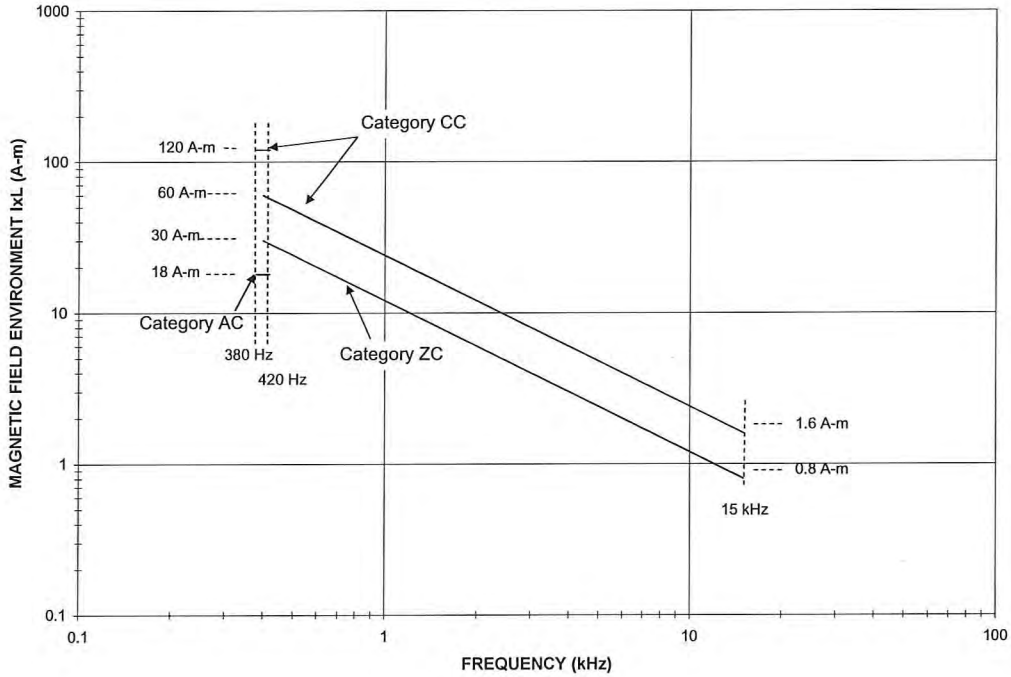

Figure 19-1(a) Audio Frequency Magnetic Field Susceptibility Test Levels – Constant Frequency

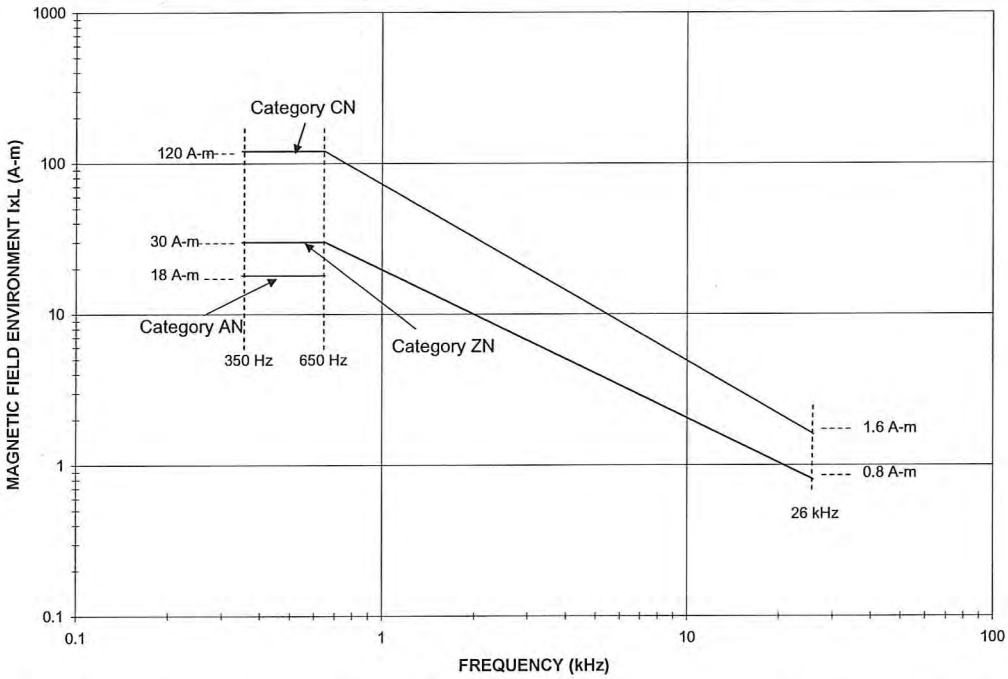

Figure 19-1(b) Audio Frequency Magnetic Field Susceptibility Test Levels – Narrow Frequency

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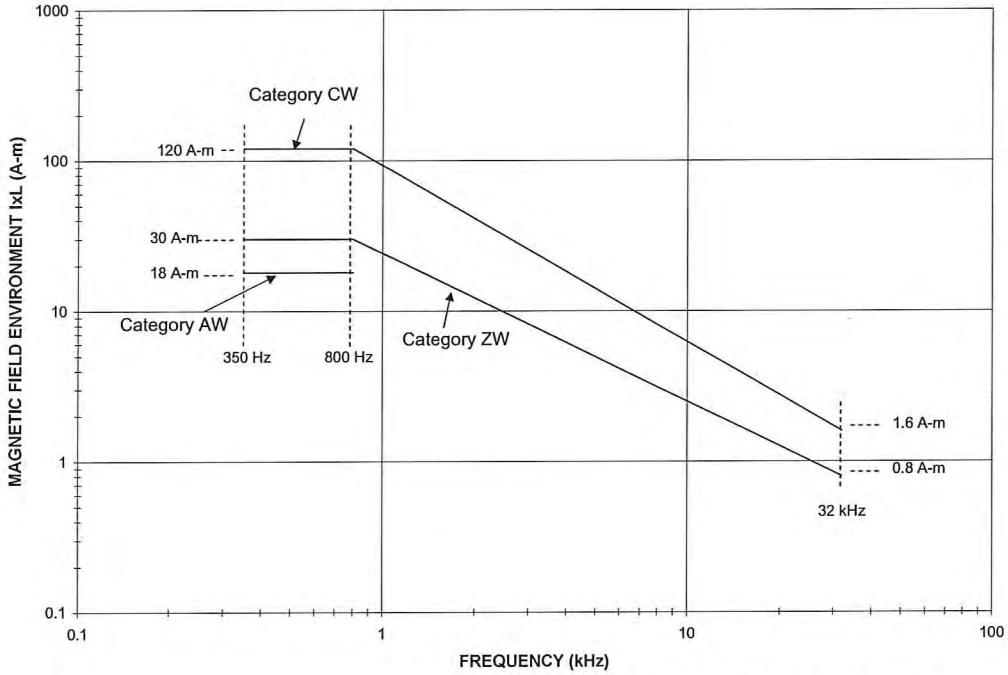

Figure 19-1(c) Audio Frequency Magnetic Field Susceptibility Test Levels – Wide Frequency

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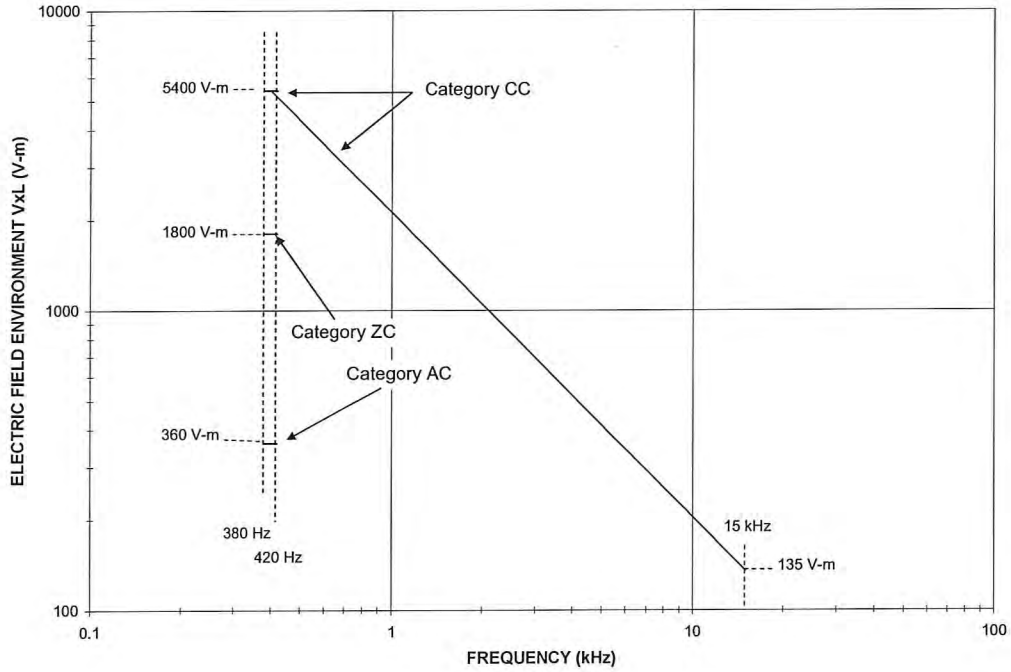

Figure 19-1(d) Audio Frequency Electric Field Susceptibility Test Levels – Constant Frequency

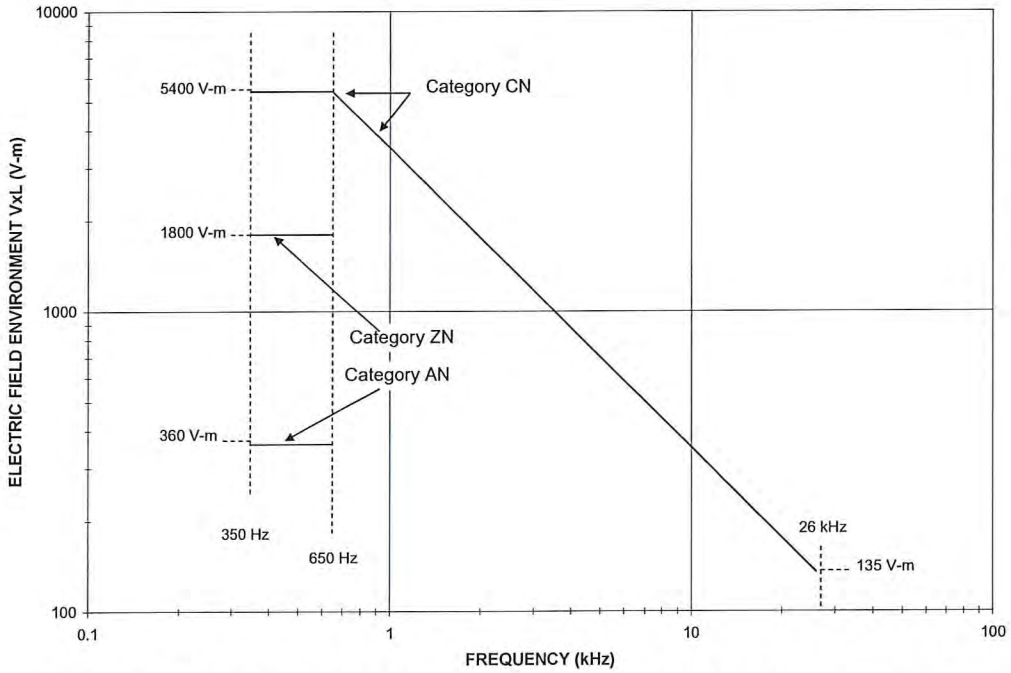

Figure 19-1(e) Audio Frequency Electric Field Susceptibility Test Levels – Narrow Frequency

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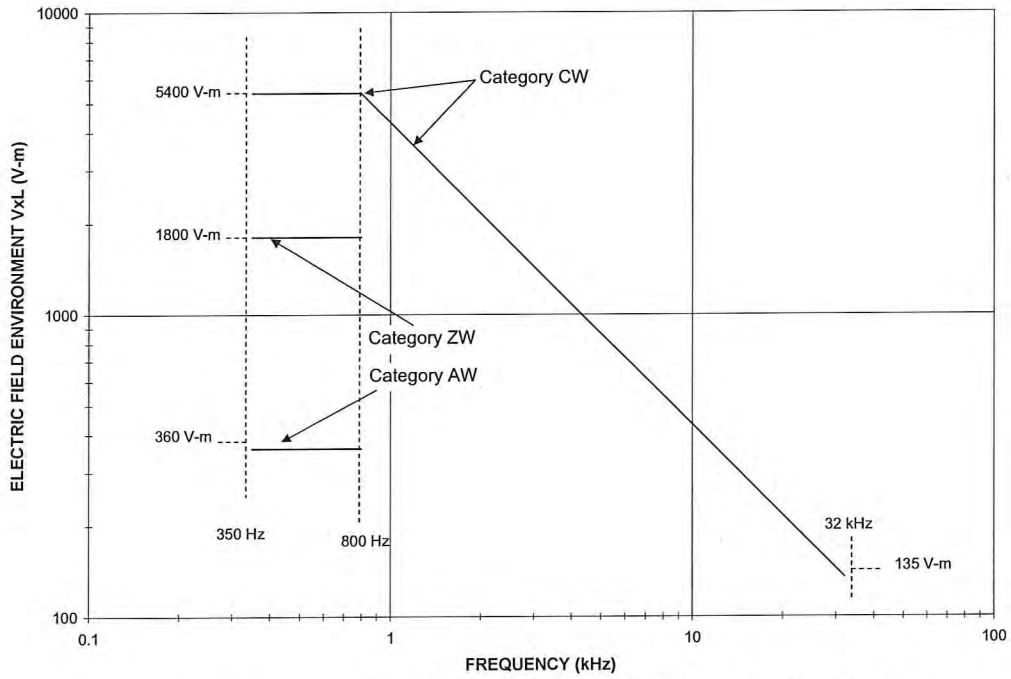

Figure 19-1(f) Audio Frequency Electric Field Susceptibility Test Levels – Wide Frequency

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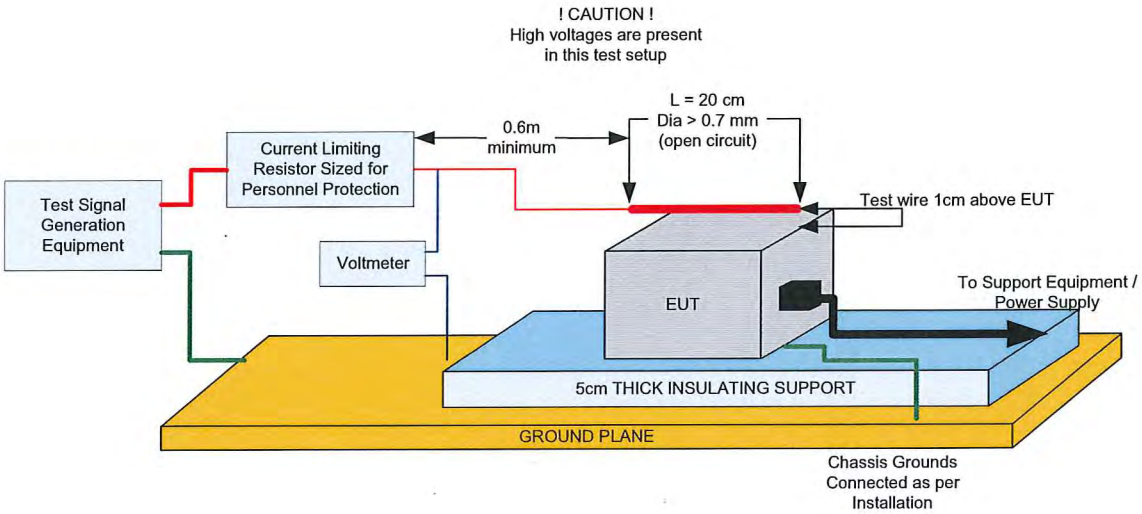

Figure 19-2 **Audio Frequency Electric Field into Equipment Susceptibility Test Setup**

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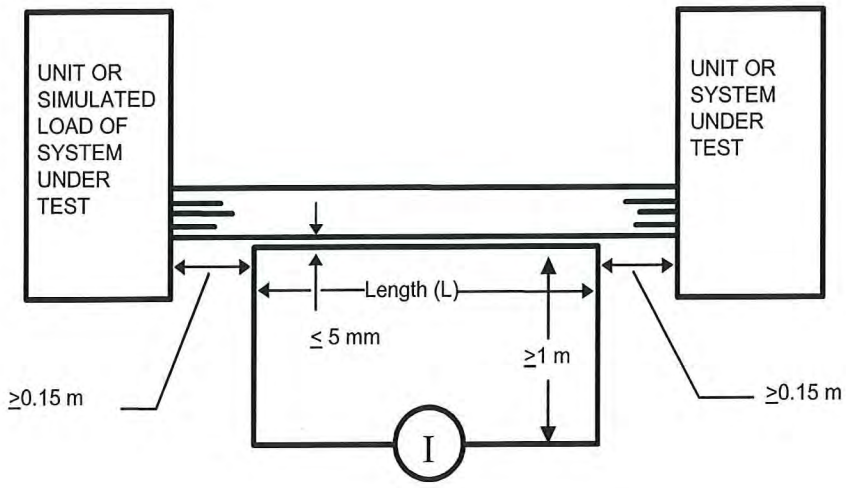

Note 1: The interconnecting wire bundle shall be spaced a minimum of 50 mm above the ground plane.

Note 2: Magnetic Field Environment = Current (I) x Length (L) (amperes rms x meters)

Figure 19-3 **Audio Frequency Magnetic Field Susceptibility Test Setup**

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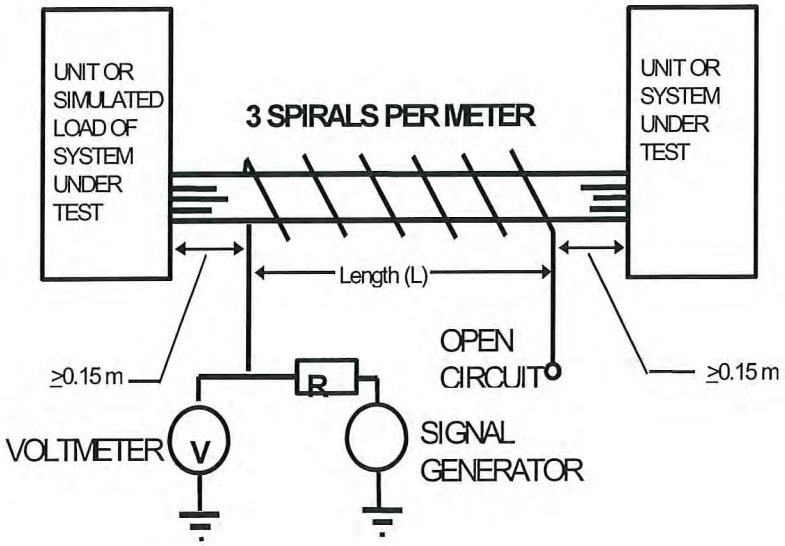

Note 1: The interconnecting wire bundle shall be spaced a minimum of 50 mm above the ground plane.

Note 2: Electric Field Environment = Voltage (V) x Length (L) (volts rms x meters)

Note 3: R sized for personnel high voltage protection

Figure 19-4 **Audio Frequency Electric Field Susceptibility Test Setup**

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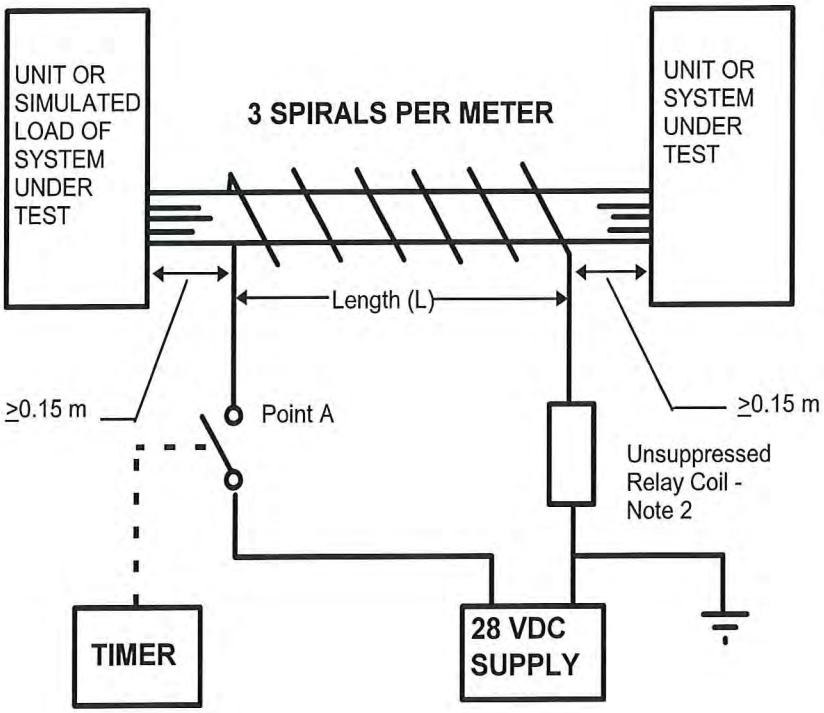

Note 1: The interconnecting wire bundle shall be spaced a minimum of 50 mm above the ground plane.

Note 2: The unsuppressed relay coil characteristics are as follows:

Voltage	=	28 volts dc
Current	=	160 mA
Resistance	=	175 ohms ± 10 %
Inductance	=	1.5 henries ± 10% in the energized position.

Note 3: 28 VDC supplied from ungrounded source with polarity reversing switch.

Figure 19-5 **Interconnecting Cable Spike Test Setup**

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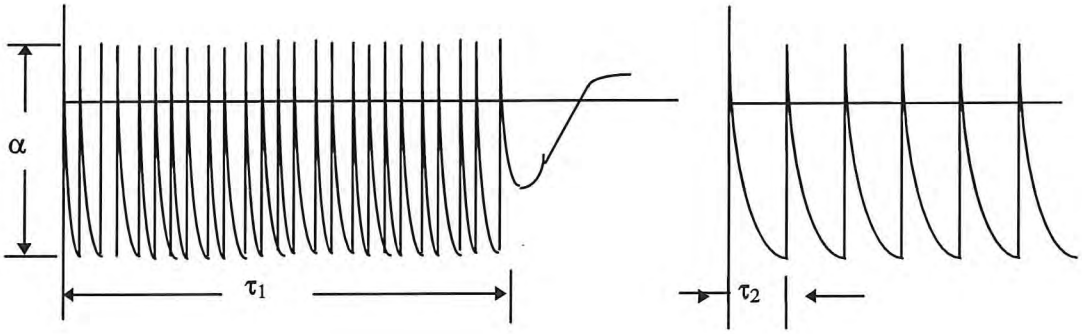

α	Amplitude $\geq$ 600 v p-p
τ_1	Total Duration 50 to 1000 microseconds
τ_2	Repetition Period 0.2 to 10 microseconds

Note: Voltage waveforms measured between Point A of Figure 19-4 and the ground plane. Inherently, the test will produce spikes of variable amplitude within a pulse. Some of the spikes may be less than the 600 V p-p level

Figure 19-6 **Inductive Switching Transients**

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